

Introductory Networking

A complete, beginner-friendly guide with every task, question, explanation and answer.

Learning Path: Network Fundamentals / Pre-Security

Difficulty: Easy | Topics: OSI Model, TCP/IP, Encapsulation, ping, traceroute, whois, dig

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About this walkthrough

This document walks through every task of the TryHackMe room **Introductory Networking**. Each task includes the concept being taught, the exact questions asked, a short explanation of how to reach the answer, and the answer itself.

What you will learn

- How the OSI model describes networking in seven layers.
- How data is encapsulated and de-encapsulated as it travels.
- How the four-layer TCP/IP model maps onto the OSI model.
- Practical use of ping, traceroute, whois and dig.

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Before you start

Try each task yourself before reading the answers - that is where the real learning happens. Several questions ask you to run commands (ping, traceroute, whois, dig) against real hosts, and answers such as IP addresses or WHOIS records can change over time. If an answer does not match, re-run the command and read the live output. All answers are shown for educational purposes.

TASK 1

Introduction

This room introduces two things at once. First, the two classic models used to describe how networks work: the **OSI model** and the **TCP/IP model**. Second, it introduces some of the most useful command-line networking tools - **ping**, **traceroute**, **whois** and **dig** - and gives you hands-on practice with each.

Key takeaway

Networking models are mental maps: they let engineers describe where something happens (a cable, an IP address, a web request) without ambiguity. The tools then let you observe those layers in action.

TASK 2

The OSI Model: An Overview

The **OSI (Open Systems Interconnection)** model describes networking using **seven layers**. Layer 1 is the physical wire carrying electrical signals; Layer 7 is the application interface you interact with. Data travels down the stack when it is sent, and back up when it is received.

Mnemonic: Please Do Not Throw Sausage Pizza Away - Physical, Data Link, Network, Transport, Session, Presentation, Application (Layer 1 to 7).

Layer	Name	Data unit	Protocols / examples
7	Application	Data	HTTP, FTP, SMTP, DNS
6	Presentation	Data	SSL/TLS, JPEG, MPEG, encryption
5	Session	Data	NetBIOS, RPC, session tracking
4	Transport	Segment (TCP) / Datagram (UDP)	TCP, UDP
3	Network	Packet	IP, ICMP, routing
2	Data Link	Frame	Ethernet, MAC addressing, PPP
1	Physical	Bits	Cables, connectors, NICs, hubs

Q1. Which layer would choose to send data over TCP or UDP?

ANSWER

4 (Transport)

Q2. Which layer checks received packets to make sure that they haven't been corrupted?

ANSWER

2 (Data Link)

Q3. In which layer would data be formatted in preparation for transmission?

ANSWER

2

Q4. Which layer transmits and receives data?

ANSWER

1 (Physical)

Q5. Which layer encrypts, compresses, or otherwise transforms the initial data to give it a standardised format?

ANSWER

6 (Presentation)

Q6. Which layer tracks communications between the host and receiving computers?**ANSWER**

5 (Session)

Q7. Which layer accepts communication requests from applications?**ANSWER**

7 (Application)

Q8. Which layer handles logical addressing?**ANSWER**

3 (Network) - logical addressing means IP addresses

Q9. When sending data over TCP, what would you call the "bite-sized" pieces of data?**ANSWER**

Segments

Q10. [Research] Which layer would the FTP protocol communicate with?**ANSWER**

7 (Application) - FTP is an application-layer protocol that runs over TCP

Q11. Which transport layer protocol would be best suited to transmit a live video?**ANSWER**

UDP - fast and connectionless; a few dropped frames do not matter

TASK 3

Encapsulation

As data moves down the stack, each layer wraps it inside a header (and, at one layer, a trailer) - like a Russian nesting doll. This is **encapsulation**. At the receiving end the process runs in reverse and is called **de-encapsulation**.

```
Application data
-> TCP header + data      = Segment
-> IP header  + segment   = Packet
-> Eth header + packet + trailer = Frame
-> transmitted as bits on the wire
```

So an Ethernet **frame** contains an IP **packet**, which contains a TCP **segment**. Only the Data Link layer adds a **trailer** (the frame check sequence used for error detection).

Q1. How would you refer to data at layer 2 of the encapsulation process (with the OSI model)?

ANSWER

Frames

Q2. How would you refer to data at layer 4 of the encapsulation process (with the OSI model), if the UDP protocol has been selected?

ANSWER

Datagrams (UDP) - TCP at layer 4 would be segments

Q3. What process would a computer perform on a received message?

ANSWER

De-encapsulation

Q4. Which is the only layer of the OSI model to add a trailer during encapsulation?

ANSWER

Data Link (layer 2)

Q5. Does encapsulation provide an extra layer of security (Aye/Nay)?

ANSWER

Aye

TASK 4

The TCP/IP Model

The OSI model is excellent for learning, but the Internet actually runs on the simpler **TCP/IP model**, which has **four layers**. The two lowest OSI layers (Physical + Data Link) collapse into **Network Interface**; the three highest (Session + Presentation + Application) collapse into **Application**.

TCP/IP layer	Covers OSI layers	Function
Application	7, 6, 5	User-facing protocols (HTTP, FTP, DNS)
Transport	4	TCP / UDP - end-to-end delivery
Internet	3	IP addressing and routing between networks
Network Interface	2, 1	Frames, MAC addresses and the physical link

TCP and the three-way handshake

TCP is a **connection-based** protocol: before any data flows, the two hosts establish a session with the three-way handshake, and flags in the TCP header track the state of that session.

Flag	Meaning	Role
SYN	Synchronise	Requests a session (step 1)
ACK	Acknowledgement	Confirms data/handshake was received (step 3)
FIN	Finish	Requests orderly termination
RST	Reset	Abruptly aborts the connection

```
Client --- SYN      ---> Server  (1) request a session
Client <-- SYN/ACK   ---  Server  (2) server agrees
Client --- ACK      ---> Server  (3) session established
```

Q1. Which model was introduced first, OSI or TCP/IP?

ANSWER

TCP/IP

Q2. Which layer of the TCP/IP model covers the functionality of the Transport layer of the OSI model (Full Name)?

ANSWER

Transport

Q3. Which layer of the TCP/IP model covers the functionality of the Session layer of the OSI model (Full Name)?

ANSWER

Application

Q4. The Network Interface layer of the TCP/IP model covers the functionality of two layers in the OSI model. These layers are Data Link, and?.. (Full Name)?

ANSWER

Physical

Q5. Which layer of the TCP/IP model handles the functionality of the OSI network layer?

ANSWER

Internet

Q6. What kind of protocol is TCP?

ANSWER

Connection-based (connection-oriented)

Q7. What is SYN short for?

ANSWER

Synchronise

Q8. What is the second step of the three way handshake?

ANSWER

SYN/ACK

Q9. What is the short name for the "Acknowledgement" segment in the three-way handshake?

ANSWER

ACK

TASK 5

Networking Tools: Ping

Ping tests whether a host is reachable and measures the round-trip time of the messages. The basic syntax is `ping <target>`, where the target can be a hostname or an IP address. Run with no count, it pings forever - stop it with Ctrl+C.

```
ping bbc.co.uk          # continuous
ping -c 3 bbc.co.uk     # -c = send 3 packets then stop
ping 8.8.8.8            # ping a host by IP address
```

Handy trick

Switches can be looked up without leaving the terminal: `man ping | grep interval`, or search the man page for a keyword with `grep`.

Q1. What command would you use to ping the bbc.co.uk website?

ANSWER

`ping bbc.co.uk`

Q2. Ping muirlandoracle.co.uk. What is the IPv4 address?

ANSWER

217.160.0.152

Q3. What switch lets you change the interval of sent ping requests?

ANSWER

`-i`

Q4. What switch would allow you to restrict requests to IPv4?

ANSWER

`-4`

Q5. What switch would give you a more verbose output?

ANSWER

`-v`

TASK 6

Networking Tools: Traceroute

Traceroute shows the path (the series of routers) a packet takes to reach a destination, and how long each hop takes. It is useful for spotting where connectivity fails, where latency spikes, or whether there is a routing loop.

```
traceroute tryhackme.com      # Linux / macOS
tracert tryhackme.com         # Windows
traceroute -i eth0 host       # -i = choose the interface
traceroute -T host            # -T = use TCP SYN requests
```

Q1. Use traceroute on tryhackme.com. Can you see the path your request has taken?

ANSWER

No answer needed - just view the hops

Q2. What switch would you use to specify an interface when using Traceroute?

ANSWER

-i

Q3. What switch would you use if you wanted to use TCP SYN requests when tracing the route?

ANSWER

-T

Q4. [Lateral Thinking] Which layer of the TCP/IP model will traceroute run on by default (Windows)?

ANSWER

Internet (layer 3 in the TCP/IP model)

TASK 7

Networking Tools: WHOIS

WHOIS returns public registration information about a domain - who registered it, the registrant's address and contact details, and important dates. It is a common first step in reconnaissance against a target. It can be run from the command line or via web tools.

```
whois facebook.com
whois microsoft.com
whois facebook.com | grep -i postal    # filter the output
```

Q1. Perform a whois search on facebook.com

ANSWER

No answer needed - run the command

Q2. What is the registrant postal code for facebook.com?

ANSWER

94025

Q3. When was the facebook.com domain first registered (Format: DD/MM/YYYY)?

ANSWER

29/03/1997

Q4. Perform a whois search on microsoft.com

ANSWER

No answer needed - run the command

Q5. Which city is the registrant based in?

ANSWER

Redmond

Q6. [OSINT] What is the name of the golf course that is near the registrant address for microsoft.com?

ANSWER

Bellevue Golf Course

Q7. What is the registered Tech Email for microsoft.com?

ANSWER

msnhst@microsoft.com (see also the Abuse / Tech contact fields)

TASK 8

Networking Tools: Dig

DNS (Domain Name System) translates human-friendly domain names into IP addresses. **dig** queries DNS servers and shows the records they return.

When you look up a domain, your computer first checks its own **local cache**. If there is no entry it queries a **recursive** DNS server, which (if it does not already know the answer) consults **root** name servers and then the relevant **Top-Level Domain** server (for .com, .co.uk and so on).

```
dig tryhackme.com

;; ANSWER SECTION:
tryhackme.com.      86400    IN      A       104.26.x.x
                   ^ TTL in seconds (86400 = 24 hours)
```

Q1. What is DNS short for?

ANSWER

Domain Name System

Q2. What is the first type of DNS server your computer would query when you search for a domain?

ANSWER

Recursive

Q3. What type of DNS server contains records specific to domain extensions (i.e. .com, .co.uk, etc)? Use the long version of the name.

ANSWER

Top-Level Domain

Q4. Where is the very first place your computer would look to find the IP address of a domain?

ANSWER

Host file

Q5. [Research] Google runs two public DNS servers. One of them can be queried with the IP 8.8.8.8, what is the IP address of the other one?

ANSWER

8.8.4.4

Q6. If a DNS query has a TTL of 24 hours, what number would the dig query show?

ANSWER

86400 (seconds in 24 hours: $24 \times 60 \times 60$)

TASK 9

Further Reading

The room closes with pointers for continuing your study of networking and the tools you just used. There are no questions to answer here.

Keep practising

Re-run `ping`, `tracert`, `whois` and `dig` against domains you know, and read their man pages (`man dig`). Seeing the output for yourself is what makes these tools stick.

Conclusion

This room covered the theory that underpins everything else in networking: the seven-layer **OSI model**, how data is **encapsulated** as it descends the stack, and the four-layer **TCP/IP model** that the Internet actually uses - including TCP's connection-based three-way handshake. It then introduced four everyday tools: **ping** for reachability, **traceroute** for path discovery, **whois** for domain registration recon, and **dig** for querying DNS.

Answer key at a glance

Task	Question	Answer
2	Layer choosing TCP or UDP	4
2	Layer checking packets for corruption	2
2	Layer formatting data for transmission	2
2	Layer that transmits and receives	1
2	Layer that encrypts/compresses/transforms	6
2	Layer that tracks communications	5
2	Layer accepting requests from applications	7
2	Layer that handles logical addressing	3
2	Bite-sized chunks of TCP data	Segments
2	FTP's layer	7
2	Best protocol for live video	UDP
3	Data at layer 2	Frames
3	Data at layer 4 over UDP	Datagrams
3	Process on a received message	De-encapsulation
3	Only layer to add a trailer	Data Link
3	Encapsulation = extra security?	Aye
4	Model introduced first	TCP/IP
4	TCP/IP layer for OSI Transport	Transport
4	TCP/IP layer for OSI Session	Application
4	Network Interface also covers OSI...	Physical
4	TCP/IP layer for OSI Network	Internet
4	Kind of protocol TCP is	Connection-based
4	What SYN is short for	Synchronise
4	Second step of the handshake	SYN/ACK
4	Short name for Acknowledgement	ACK
5	Ping bbc.co.uk	ping bbc.co.uk
5	muirlandoracle.co.uk IPv4	217.160.0.152
5	Change ping interval	-i
5	Restrict ping to IPv4	-4

Task	Question	Answer
5	More verbose ping	-v
6	Traceroute interface switch	-i
6	Traceroute TCP SYN switch	-T
6	Traceroute's TCP/IP layer (Windows)	Internet
7	facebook.com registrant postal code	94025
7	facebook.com first registered	29/03/1997
7	microsoft.com registrant city	Redmond
7	Golf course near Microsoft	Bellevue Golf Course
7	microsoft.com tech email	msnhst@microsoft.com
8	DNS short for	Domain Name System
8	First server your computer queries	Recursive
8	Server for .com / .co.uk records	Top-Level Domain
8	First place your computer looks	Host File
8	Google's other public DNS	8.8.4.4
8	TTL of 24 hours shown by dig	86400

Next steps

Move on to the **OSI Model** and **Packets & Frames** rooms to go deeper on layers 1-3, and revisit the tools here whenever you troubleshoot a connection. Remember that WHOIS and DNS answers change over time, so always verify against live output.